SP25 Vorobeychik Independent Study

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1 Introduction

My name is Nasir Sims, and I am a second year Computer Science student at Washington University in St. Louis. I am continuing my second semester working with Dr. Vorobeychik on his research to advance autonomous driving systems. I am continuing my research into intersection handling and precise maneuvers through the open-source CARLA environment, and the 1:8 scale "Mini-City" environment.

2 Code Basis

Last semester, I had worked closely with Owen Ma and his thesis work, V-E2E-RL-AD to build a reinforcement learning model for intersection handling. This model utilized Carla and Pytorch to train an ego car to stay in-between the lane-lines in multiple different city environments. I found that getting this repository running worked well, however I had a hard time understanding the scope and adding to the codebase as I did not help build it. ChatGPT could only take me so far, so I employed the help of a previous researcher, Caitlyn Johnson to help gain a better understanding of the work that she did regarding her work on intersection handling.

2.1 Caitlyn

Early in the semester, I sent an email to Caitlyn Johnson, a researcher who also forked Owen's repository to develop her own model and present different findings. Last summer, she worked on creating a balanced dataset of 700 images of left and right turn images generated from different Carla town environments. She also used the forked repository of v-e2e-rl-ad to do something with these images in code. Lane detectors were added, LaneNet was retrained, and a new high-level class was added for alternative route planning.

3 Classification vs. Reinforcement Learning

Upon my meeting with Owen at the beginning of the year, I was given a choice: continue to develop v-e2e-rd-ad in the reinforcement learning approach, or build a classification model from scratch. This choice was difficult because my understanding in machine learning has been a "learn-as-you-go" approach, so the idea of learning a new type of model was daunting. However at that time I felt that I was hitting a roadblock in my code, as I felt I did not complete the work I wanted to by the end of my first semester, so I chose to take the classification model approach.

4 Classification & Images

4.1 Classification Model

A classification model is a type of supervised machine learning algorithm designed to assign inputs into predefined, discrete categories based on learned patterns from labeled training data. In the context of my project, I tried to build my own classification model to support decision-making at intersections. Specifically, the model was trained to classify front-facing camera images into one of two maneuver categories: left turn or right turn, utilizing a CARLA dataset each annotated with the correct maneuver label. The visual input data was preprocessed and passed through a convolutional neural network (CNN), which learned to identify spatial features and patterns indicative of upcoming left or right turns—such as lane geometry, road markings, and surrounding environmental cues.

Once trained, this model was integrated into the driving loop as a high-level policy module. At each timestep, the simulator feeds the current windshield image to the classifier, which predicts whether a left or right turn is required. This prediction is then passed to a separate control module responsible for executing the appropriate low-level vehicle control commands, such as applying a fixed steering angle over a sequence of frames or engaging a PID controller. By decoupling perception (via classification) from control, the system enables interpretable and modular behavior planning, allowing for more robust intersection handling in both scripted and dynamic driving scenarios.

4.2 Image Dataset

From my talk with Caitlyn I obtained the results of here REU Wrapup, which most importantly is her dataset. It includes the right turn dataset with 359 files, and the left turn dataset with 330 files, as well as an accompanying csv file with their trained RL models.

I am very glad that I reached out to Caitlyn because without this, I was going to obtain my windshield image dataset via the photos captured from the front-facing camera on the F1Tenth vehicle platform placed in lanes and intersections in the mini-city environment. This was utilized to train Owen's v-e2e-rl-ad RL model, and while it would have worked fine, there were some limitations such as: resizing issues, quality of images, and number of images. By manually selecting the best images used to train my classification model, it would have limited my scope because it would have taken too long to create a new dataset using the F1Tenth vehicle, so it limited the number of images.

Below are figures comparing two similar images in the different datasets, labeled accordingly.

(a) Left Intersection (b) Right Intersection

Figure 1: Owen Ma's F1Tenth Dataset

(a) Left Intersection (b) Right Intersection

Figure 2: Caitlyn Johnson's CARLA Dataset

5 Code Changes

With the appropriate dataset, it was now time to transform the existing code to align with a new classification model instead of reinforcement learning.

5.1 nv-e2e-cl-ad Repository

To start this project, I forked the repository from Caitlyn and named it to 'nv-e2e-cl-ad' to reference the previous nomenclature, changing the 'verification' and 'classification' status in the title. From here is where I did the primary code editing.

5.2 train_ppo.py

This method is used to set baselines for training the classification model:

```
def train_model(data_path, csv_path, epochs=10, batch_size=32, lr=1e-4):
    transform = transforms.Compose([
        transforms.Resize((256, 256)),
        transforms.ToTensor()
    ])
    dataset = TurnDataset(csv_file=csv_path, img_dir=data_path, transform=transform)
    dataloader = DataLoader(dataset, batch_size=batch_size, shuffle=True)
```

This method is used for iterating the epochs in the classification model:

```
        for epoch in range(epochs):
        running_loss = 0.0
        correct = 0
        total = 0
        for inputs, labels in tqdm(dataloader, desc=f"Epoch {epoch+1}/{epochs}"):
            inputs, labels = inputs.to(DEVICE), labels.to(DEVICE)
            optimizer.zero_grad()
            outputs = model(inputs)
            loss = criterion(outputs, labels)
            loss.backward()
            optimizer.step()
```

5.3 alt_global_route_planner.py

This file was created by Caitlyn and added onto the original LaneNet implementation of route_planner.py, but with changes to incorporate intersection handling.

5.4 PID

After making the appropriate changes to the codebase, it came PID. Owen had coded the LaneNet reinforcement learning model appropriately, but I had no idea what to do when it came to hard-coding these constants for steering control. In my code I did not get it working properly. I believe that my implementation for incorporating them into the correct epochs was incorrect.

6 Limitations & Future Steps

6.1 Limitations

This semester presented some different challenges in my continued work in this independent study. Some limitations that I encountered stemmed from tackling a new learning process when it came to learning how to write my classification model. When it came to starting in this lab, I knew my limited knowledge in machine learning would be difficult, and learning yet another model and implementing it in code took up a significant amount of time.

The flooding of the McKelvey building also made it challenging to work on my research, as I do my work on the code on the physical PC owned by Dr. Vorobeychik on the 3rd floor PhD office, however I simply used that time to work on my report.

6.2 Future Steps

Unfortunately, this will be my final semester with Dr. Vorobeychik's lab. I was extremely lucky to have stumbled upon Dr. Vorobeychik's lab by accident, and I have learned so much so quickly due to the weekly meetings, and the team, especially Owen being so readily available to meet and answer questions. However I think that my lack of a proper introduction to machine learning topics made the learning processes feel like an uphill battle. Conversations with Owen were sometimes lost on me when it came high-level topics, and resources such as ChatGPT did not make me feel like I was learning in a way that was satisfying.

7 Conclusion

Throughout this school year, this independent study has provided me with so many learning opportunities that I would not have been able to realize until much later in my CS journey. I have been able to rapidly research many topics related to machine learning through my intersection handling project. Despite not being in a place where I can represent data via CARLA in a presentable way, I have gained so much from this Independent Study in terms of communication, coding ability, and learning.

8 References

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